

CHAVALI UDDIPA DATTUDU

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PROFESSIONAL SUMMARY

Electronics and Communication Engineering student focusing on the VLSI domain. Gaining foundational knowledge in RFIC design and analog-to-digital converters through academic research internships. Capable of building, simulating, and designing custom layouts for hardware using Cadence Virtuoso and Verilog.

ACADEMIC BACKGROUND

Madanapalle Institute of Technology and Science

Bachelor of Technology in Electronics and Communication Engineering; CGPA: 9.37

Andhra Pradesh, India

July 2025 – Expected May 2028

- **Relevant Coursework:** Analog Circuits, Digital System Design (Verilog HDL), Electronic Devices & Circuits, Network Theory, Python Programming, PCB Design.

ESC Government Polytechnic College

Diploma in Electronics and Communication Engineering; Percentage: 95.2%

Andhra Pradesh, India

Aug 2022 – May 2025

TECHNICAL SKILLS

Core Competencies: Analog Circuit Design, Radio Frequency CMOS Integrated Circuit Design, DC/AC/Transient Simulations.

EDA Tools: Cadence Virtuoso, Altium Designer (Basic PCB), NGSPICE, NI Multisim, MATLAB.

Programming & HDLs: Verilog HDL, Python (Currently learning Data Science & Basic Automation), C language.

INTERNSHIPS & EXPERIENCE

National Institute of Technology (NIT) Calicut

Research Intern - Analog IC Design for Image Sensors

Remote

July 2026 – Present

- Collaborating on remote research under the supervision of Dr. Bhuvan B, focusing on the study and design of Programmable Gain Amplifiers (PGAs) and Single-Slope ADCs for CMOS camera image sensors.
- Working on the transistor-level modeling and simulation of the converter blocks to optimize gain, noise, and power for high-resolution imaging applications.

National Institute of Technology (NIT) Calicut

Summer Research Intern – RFIC Design for Anti-Drone Radar

Calicut, Kerala

May 2026 – July 2026

- Designed and simulated an unconditionally stable 9.0 GHz cascode Low Noise Amplifier (LNA) in a 180nm CMOS process using Cadence Virtuoso under the guidance of Dr. Karthik R, achieving a forward voltage gain (S_{21}) of 11.06 dB and a Noise Figure (NF) of 1.41 dB.
- Engineered an active LC L-match output network and multi-finger gate topologies (30 fingers of $5.0\ \mu\text{m}$) to neutralize the Miller effect, ensuring precision $50\ \Omega$ impedance matching ($S_{11} = -17.06\ \text{dB}$, $S_{22} = -23.27\ \text{dB}$).
- Verified high power efficiency via ADE L DC operating point extraction, yielding a low power consumption (P_{DC}) of 14.13 mW and an exceptional Figure of Merit (FoM) of 21.05 GHz/mW for X-band receiver front-ends.

ENGINEERING PROJECTS

Automated CMOS Inverter Characterization | Python, NGSPICE, Pandas

July 2026 – Aug 2026

- Wrote a Python automation script that runs NGSPICE silently in the background, automatically sweeping 180nm CMOS transistor widths from $1.0\ \mu\text{m}$ to $5.0\ \mu\text{m}$. Used Pandas to extract and parse the raw simulation data directly from text files, replacing manual GUI testing to efficiently analyze Voltage Transfer Characteristics (VTC).
- Graphed the shifting switching thresholds using Matplotlib to visually prove how changing the PMOS size alters the strength of the pull-up network.

180nm 6T SRAM Cell Design and Characterization | Analog VLSI Design

June 2026 – July 2026

- Designed a 6T SRAM cell in a 180nm CMOS process using Cadence Virtuoso, optimizing transistor sizing (Cell Ratio = 2.0, Pull-Up Ratio = 0.7) for balanced read/write stability.
- Extracted Voltage Transfer Characteristics (VTC) and applied Seevinck's method via a custom MATLAB script to calculate a robust Read Static Noise Margin (SNM) of 309.0 mV.
- Performed transient and DC analyses to validate energy efficiency, achieving an active dynamic power of $86.39\ \mu\text{W}$ (at a 40ns period) and an ultra-low static leakage of 24.1 pW.

HONORS & AWARDS

2nd Position, "Circuit Detective" | ECLECTICA 2K26 National Symposium, MITS

April 2026

1st Prize, "Circuit Hunt" & Tech Quiz | E-Merge 2K24 National Symposium, JNTUACEA

Nov 2024

1st Prize, Paper Presentation | Ripple 2K24 PRAVAHA, RGM College

March 2024

Academic Excellence Awards | Top performance across 3 Diploma semesters

2023 – 2025